ENHANCING COVERT COMMUNICATION: WHISPERING SECRETS IN A CROWD

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ABSTRACT

Consider a Multi-Access communication setup where a subset of users need to be covert, i.e. their communication should be undetectable to an external warden.
In this work:

- We derive a **full characterization** of the achievable rates.
- We show that the presence of the non-covert users allows us to **reduce the secret-key rate** and **increase the message rate** of the covert users.

SETUP OF COMMUNICATION

Define the message and key sets

$$\mathcal{M}_1 \triangleq \{1,\dots,M_1\}, \quad \mathcal{M}_2 \triangleq \{1,\dots,M_2\}, \quad \mathcal{K} \triangleq \{1,\dots,K\}, \tag{1}$$

for given numbers M_1, M_2, and K and let the messages W_1 and W_2 and the key S be uniform over $\mathcal{M}_1, \mathcal{M}_2$, and $\mathcal{K}$ respectively.

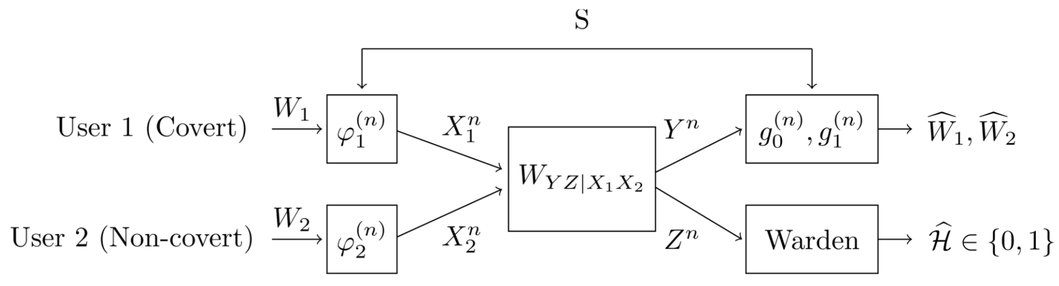


Fig. 1: Multi-access communication where communication of User 1 has to remain undetectable to an external warden.

Reliability:

$$P_{e0} \triangleq \Pr\Big(\widehat{W}_2 \neq W_2 \Big| \mathcal{H}=0\Big). \tag{2}$$

$$P_{e1} \triangleq \Pr\Big(\widehat{W}_2 \neq W_2 \text{ or } \widehat{W}_1 \neq W_1 \Big| \mathcal{H}=1\Big) \tag{3}$$

Covertness:
For each $w_2 \in \mathcal{M}_2$ and $W_2 = w_2$, define the warden's output distribution under $\mathcal{H}=1$

$$\widehat{Q}^n_{\mathcal{C},w_2}(z^n) \triangleq \frac{1}{M_1 K} \sum_{(w_1,s)} W^{\otimes n}_{Z|X_1X_2}(z^n|x_1^n(w_1,s), x_2^n(w_2)), \tag{4}$$

and under $\mathcal{H}=0$

$$W^{\otimes n}_{Z|X_1X_2}(z^n|0^n, x_2^n(w_2)), \tag{5}$$

and the divergence (distance) between these two distributions:

$$\delta_{n,w_2} \triangleq \mathbb{D}\Big(\widehat{Q}^n_{\mathcal{C},w_2} \Big\| W^{\otimes n}_{Z|X_1X_2}(\cdot|0^n, x_2^n(w_2))\Big), \quad w_2 \in \mathcal{M}_2. \tag{6}$$

Requirements on Codes:

$$\lim_{n\to\infty} \delta_{n,w_2} = 0, \qquad w_2 \in \mathcal{M}_2, \tag{7}$$

$$\lim_{n\to\infty} P_{ei} = 0, \qquad i \in \{0,1\}. \tag{8}$$

Coding scheme:
Fix a large blocklength n and define a pmf P_T and two conditional pmfs $P_{X_{1,n}|T}$ and $P_{X_2|T}$.

Encoding: Communicate in two phases. In **Phase 1**: Use codebooks of statistics $P^{(1)}_{X_{1,n}}$ and $P^{(1)}_{X_2}$. In **Phase 2**: Use codebooks of statistics $P^{(2)}_{X_{1,n}}$ and $P^{(2)}_{X_2}$. See Figure 2.

Decoding: Successive decoding. First decode W_2 then W_1. See Figure 3.

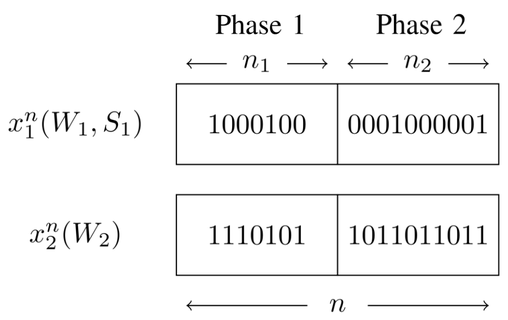


Fig. 2: Encoding procedure

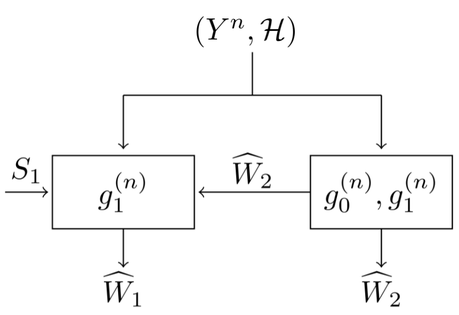


Fig. 3: Decoding procedure

MAIN RESULT

THEOREM 1

There exists a sequence of encoding and decoding functions satisfying (7)-(8) **if, and only if,**

$$r_2 \triangleq \lim_{n\to\infty} \frac{\log(M_2)}{n} \leq \mathbb{I}(X_2; Y \mid X_1 = 0, T), \tag{9}$$

$$r_1 \triangleq \lim_{n\to\infty} \frac{\log(M_1)}{\sqrt{n\mathbb{E}_{W_2}[\delta_{n,W_2}]}} \leq \sqrt{2}\frac{\mathbb{E}_{P_{TX_2}}[\epsilon_T D_Y(X_2)]}{\sqrt{\mathbb{E}_{P_{TX_2}}[\epsilon_T^2 \cdot \chi_{2,Z}(X_2)]}}, \tag{10}$$

$$k \triangleq \lim_{n\to\infty} \frac{\log(K)}{\sqrt{n\mathbb{E}_{W_2}[\delta_{n,W_2}]}} \geq \sqrt{2}\frac{\mathbb{E}_{P_{TX_2}}[\epsilon_T (D_Z(X_2) - D_Y(X_2))]}{\sqrt{\mathbb{E}_{P_{TX_2}}[\epsilon_T^2 \cdot \chi_{2,Z}(X_2)]}}, \tag{11}$$

for some pmf P_{TX_2} over $\mathcal{T} \times \mathcal{X}_2$ and $\epsilon_1, \epsilon_2 \in [0,1]$.
Here, we define for any $x_2 \in \mathcal{X}_2$:

$$D_Y(x_2) \triangleq \mathbb{D}\big(W_{Y|X_1X_2}(\cdot|1,x_2) \,||\, W_{Y|X_1X_2}(\cdot|0,x_2)\big), \tag{12}$$

$$D_Z(x_2) \triangleq \mathbb{D}\big(W_{Z|X_1X_2}(\cdot|1,x_2) \,||\, W_{Z|X_1X_2}(\cdot|0,x_2)\big), \tag{13}$$

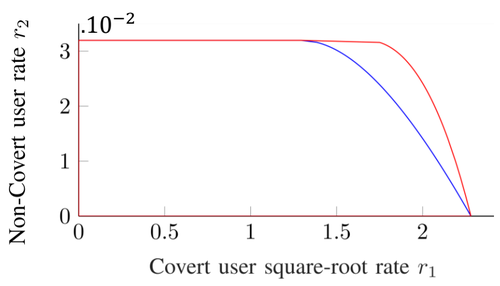
$$\chi_{2,Z}(x_2) \triangleq \chi_2\big(W_{Z|X_1X_2}(\cdot|1,x_2) \,||\, W_{Z|X_1X_2}(\cdot|0,x_2)\big), \tag{14}$$

and we recall that:

$$\mathbb{D}(P\|Q) = \sum_{x\in\mathcal{X}} P(x)\log\frac{P(x)}{Q(x)}, \quad \chi_2(P\|Q) = \sum_{x\in\mathcal{X}} \frac{(P(x)-Q(x))^2}{Q(x)}$$

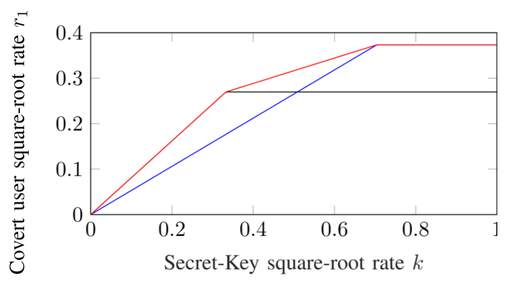
General binary input channel example:

Time-sharing improves the rates



- With two phases $|\mathcal{T}| = 2$ (in red).
- With a single phase $|\mathcal{T}| = 1$ (in blue).

Non-covert user increases capacity of covert users



- Non constant channel inputs X_2 at User 2 (in red).
- Constant channel inputs $X_2 = 0$ at User 2 (in blue).
- Constant channel inputs $X_2 = 1$ at User 2 (in black).

THE CROWD CACOPHONY

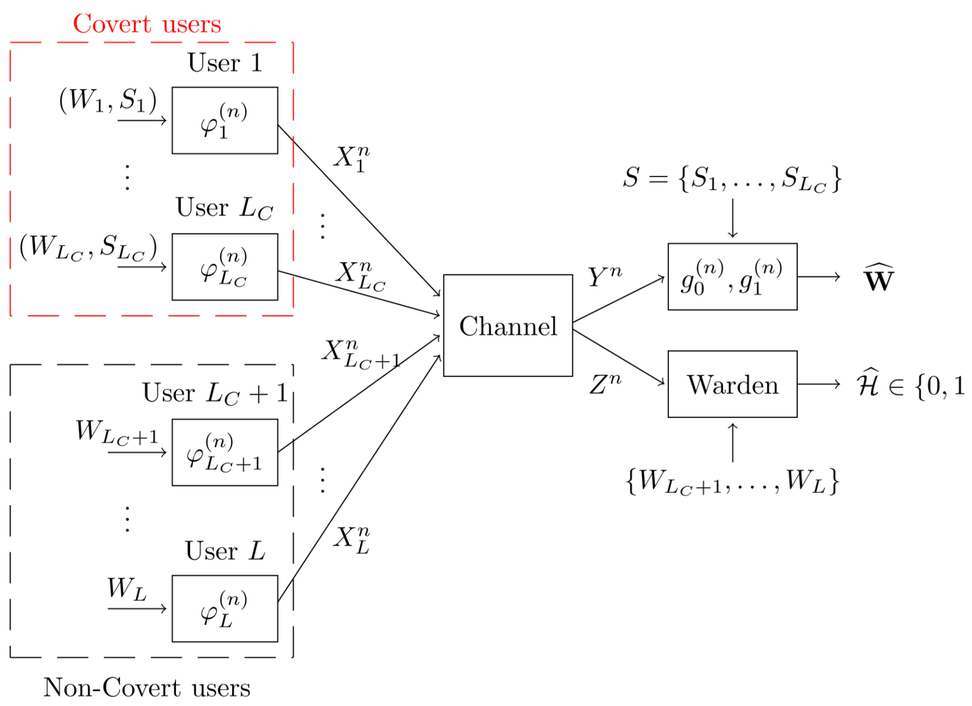


Fig. 4: Multi-Access communication setup where a subset of users need to be covert.

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